

FIZA ASLAM

ML Engineer | Predictive Maintenance, NLP & Model Deployment | Python, XGBoost, Streamlit

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SUMMARY

ML Engineer and BS IT student (7th semester) at The Islamia University of Bahawalpur, specializing in predictive maintenance, NLP, and deployed AI applications. Built and deployed production ML systems — including a predictive maintenance model with 99% accuracy and zero missed failures on 997 test cases, and a churn model deployed as a live Streamlit app. Comfortable owning a project from data ingestion through deployment.

EXPERIENCE

AI Engineer Intern — CodeCelix Jan 2026 – Apr 2026 | Remote

- Built and deployed multimodal AI models for TTS (text-to-speech), STT (speech-to-text), and OCR — processing 1,000+ unstructured files weekly
- Integrated models into production workflows using FastAPI and Docker — reducing manual data extraction time by 30% (10 hrs → 7 hrs/week)
- Delivered 5+ AI projects on time, each serving 10+ end-users in production
- Collaborated with engineering, product, and design teams to align AI solutions with business goals

Machine Learning Intern — FlyRank AI Jun 2026 – Present | Remote

- Built end-to-end ML pipelines using Python and Scikit-learn, processing 10,000+ data points daily for automated organic growth
- Trained Decision Tree and XGBoost models achieving 84.7% accuracy (precision@50), resulting in 0 false positives on high-risk predictions
- Deployed models via Streamlit, serving 5+ internal teams with real-time predictions
- Automated feature engineering, cutting manual data processing by 20+ hours/week (~100 hours/month saved for the team)

PROJECTS

EV Safety System — AI Predictive Maintenance | Python, Scikit-learn, Random Forest, Pandas, Streamlit

- Predicted EV charger failures 7 days in advance with 99% accuracy (AUC-ROC: 0.9999)
- Achieved zero missed failures (997/997 true positives) with only 20 false alarms
- Deployed interactive risk-scoring dashboard via Streamlit with color-coded maintenance alerts

Customer Churn Prediction — Telecom AI App | Python, Scikit-learn, Random Forest, Logistic Regression, Streamlit

- Built telecom churn prediction app with 78% accuracy using demographic, billing, and service data
- Applied SMOTE to correct class imbalance; deployed as a Streamlit app with confidence scores and feature importance

PriceOptimizer AI — Dynamic Pricing Engine | Python, XGBoost, Scikit-learn, Streamlit, Plotly

- Built a demand-prediction model ($R^2 = 0.88$, MAE = 5.9 units) trained on 3,000+ samples
- Optimized pricing across 80+ price points; added a What-If simulator for competitor price changes

Resume Analyzer — NLP Resume-JD Match Scoring | Python, NLTK, Sentence-Transformers (SBERT), Streamlit

- Built an NLP tool that scores resume-to-job-description fit using a 240+ term skill taxonomy plus SBERT semantic similarity
- Combined skill match (60%) and semantic similarity (40%) into a weighted score, with matched/missing skills and suggestions shown in the UI
- Found and fixed real matching bugs during testing (lemmatization/punctuation errors, false-positive matches like “typing” vs. “Job Type”) and deployed the tool live on Streamlit

SKILLS

- ML/AI:** XGBoost, Random Forest, Machine Learning, Deep Learning, NLP, Predictive Modeling, Model Deployment, Model Evaluation
- Cloud:** Microsoft Azure (Azure ML, App Service, Blob Storage, Functions)
- Libraries:** Scikit-learn, TensorFlow/Keras, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Hugging Face, Plotly
- Tools & Deployment:** Streamlit, FastAPI, Docker, Jupyter Notebook, VS Code, GitHub, Google Colab, Kaggle, Azure Portal
- Languages:** Python, SQL

EDUCATION

BS Information Technology — The Islamia University of Bahawalpur | 2023 – 2027

FSc Pre-Engineering — Abaidullah College, Haveli Lakha | 2021 – 2023

CERTIFICATIONS & LANGUAGES

- Python for Data Science, Data Analytics, Microsoft Azure Fundamentals (AZ-900 — In Progress)
- Languages: English, Urdu, Punjabi